

The .causal Format: Embedded Deterministic Inference for Domain-Agnostic Knowledge Graph Amplification

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Abstract—Every existing knowledge graph format separates storage from inference—RDF from SPARQL, Neo4j from Cypher—making inference non-portable, engine-dependent, and leaving weak signals permanently hidden below the detection threshold.

The .causal binary format eliminates this separation. A three-pass deterministic engine—exact keyword chaining, semantic direction propagation, and Jaro-Winkler fuzzy entity matching (≥ 0.85)—executes during deserialization. The same input always produces the same inferred triplets with full provenance tracking.

The key result: convergence points with only 1–3 explicit triplets become detectable (21–30 supporting chains) through transitive closure, surfacing patterns invisible to flat-table storage. Cross-domain validation across seven domains without modifying the inference engine yields $1.9\times$ to $6.8\times$ amplification, $20.8\times$ compression on genomic data (3.06 GB $\rightarrow$ 147 MB), and 497/497 correctness against independent nuclear reference data. The format transforms static records into self-amplifying knowledge.

Index Terms—binary format, data lifecycle reuse, deterministic inference, knowledge graph, signal amplification, symbolic AI, transitive closure

I. INTRODUCTION

Three triplets about vagus nerve dysfunction in a database of 5,084 causal facts look like noise—below any reasonable detection threshold. After deterministic transitive inference in the .causal format: 21 supporting chains spanning multiple papers, multiple mechanisms, multiple entry points. That is not noise. That is a convergence hub. The inference did not invent these connections—it computed what was already logically entailed by the explicit facts across 166 papers. Three more concepts followed the same pattern: POTS/autonomic axis (3 $\rightarrow$ 23, $7.7\times$), autonomic-cardiovascular link (1 $\rightarrow$ 30, $30\times$), and endothelial-mitochondrial coupling (3 $\rightarrow$ 18, $6\times$). All four were completely invisible in SQL. All four became obvious convergence hubs after transitive closure. The format made them visible without being asked.

Every existing knowledge graph format separates storage from reasoning. RDF [1] stores typed triples; OWL adds class hierarchies; SPARQL or external reasoners derive entailments. Neo4j [2] stores property graphs; Cypher queries traverse them. JSON-LD [3] serializes linked data; external engines complete it. This separation causes three failures: (1) reproducibility depends on the engine version; (2) weak signals stay hidden

unless the analyst writes the right query; (3) portability requires shipping the engine alongside the data.

The .causal format eliminates this separation. Inference rules are embedded in the binary file and execute during deserialization. Open the file—the transitive closure is computed. No engine to configure. No query to write. No analyst intuition required. The file arrives pre-amplified: every connection derivable from the data is already materialized. For an AI agent, this means the knowledge graph arrives complete—connections spanning hundreds of source documents, traceable to evidence, ready for reasoning.

We validate across seven scientific and engineering domains with zero modification to the core three-pass inference engine (Table III). Domain adaptation occurs exclusively at the extraction layer—how source data becomes triplets (detailed in the companion paper [4]). The core engine is never changed.

We make three contributions:

- 1) The first knowledge graph format where **inference executes during deserialization**—the file is the engine, not a container requiring an external reasoner process.
- 2) **Weak signal amplification**: convergence points with only 1–3 explicit triplets become detectable (21–30 supporting chains) through transitive closure, enabling discovery of patterns invisible to flat-table storage.
- 3) **Domain-agnostic validation** across seven domains—biomedicine, incident analysis, genomics, cryptanalysis, security tool synthesis, nuclear physics, and gravitational waves—with one format, one engine, and zero domain-specific inference code.

II. THE .CAUSAL FORMAT

A. Binary Structure

The file begins with an 8-byte header: a 6-byte magic sequence CAUSAL followed by a uint16 version number. The remainder is a zlib-compressed MessagePack payload containing four logical sections. All sections are serialized with MessagePack and compressed with zlib (level 6), achieving a combined 4.7:1 compression ratio on biomedical data and $20.8\times$ on genomic data.

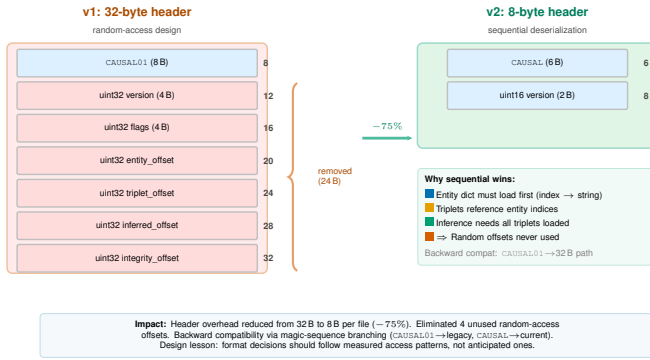


Fig. 1. Header evolution from v1 (32-byte, random-access) to v2 (8-byte, sequential). All four section-offset fields (24 bytes) were eliminated after production measurement showed sequential deserialization in 100% of use cases across seven domains. Backward compatibility is maintained via magic-sequence branching.

The initial format specification used a 32-byte header designed for random access: an 8-byte magic sequence (CAUSAL01), uint32 version, uint32 flags, and four uint32 section offsets enabling direct jumps to any section without parsing preceding data. This design assumed that consumers would need random access to individual sections—a reasonable assumption for database-like workloads. Production deployment across seven domains revealed that this assumption was wrong: all use cases deserialize sequentially, because the inference engine must process entities before triplets (to resolve indices) and triplets before inferred facts (to compute transitive closures). The current 8-byte header (6-byte magic CAUSAL, uint16 version) reduces header overhead by 75%. Backward compatibility is maintained through the magic-sequence check: files with the old CAUSAL01 magic are detected and parsed with the legacy 32-byte layout. The lesson generalizes: format design should be driven by measured access patterns, not anticipated ones.

Fig. 1 illustrates the structural difference and lists the design rationale. The change is small—24 bytes removed—but the principle is significant: four offset fields existed to support a use case (random section access) that never occurred in production across seven domains and 780K+ triplets. The evolution demonstrates that empirical measurement of actual access patterns, not anticipated workloads, should drive binary format design.

Entity dictionary. All entity strings are deduplicated into an indexed dictionary. Triplets reference entities by index, enabling $O(1)$ lookup. On genomic data, 707,689 triplets reference only 488,833 unique entity strings—the dictionary alone saves 31%.

Triplet section. Each explicit triplet stores subject index, mechanism index, object index, confidence $c \in [0, 1]$, and provenance metadata.

Inferred section. Same structure as triplets, plus a provenance array listing source triplet indices and the rule identifier. Every inferred fact is fully traceable.

Integrity. xxhash64 checksums cover each section; integrity

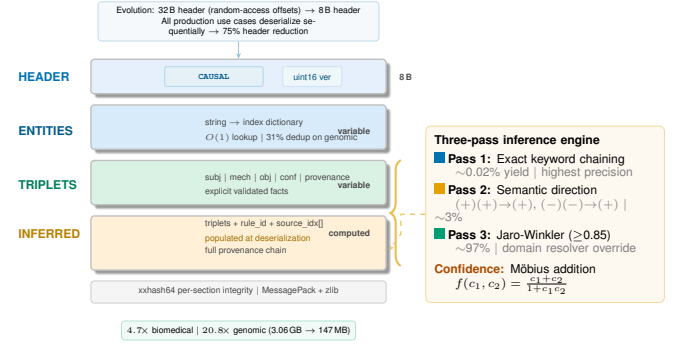


Fig. 2. Binary structure of the .causal format. The 8-byte header (evolved from a 32-byte random-access design after production measurement showed sequential deserialization in all use cases) is followed by four compressed sections. The three-pass inference engine populates the inferred section at deserialization time—the file is the engine.

is verified at load time.

B. Three-Pass Inference Engine

The engine executes during deserialization—deterministically, with zero stochastic components. Three passes apply complementary matching strategies with increasing recall at each stage.

Definition 1 (Inference Rule): Given triplets $A \xrightarrow{m_1} B$ and $B \xrightarrow{m_2} C$, and a rule R mapping $(m_1, m_2) \mapsto m_3$ with confidence modifier γ , infer $A \xrightarrow{m_3} C$ with $c = c_1 \cdot c_2 \cdot \gamma$.

Pass 1 (Exact keyword chaining): Finds chains where mechanism keywords match defined rules exactly. Highest precision, lowest yield (~0.02% of inferences in biomedical data). Complexity $O(n \times r)$.

Pass 2 (Semantic direction propagation): Mechanisms classified as positive, negative, or neutral. Direction composition follows physical logic: (+) + (+) $\rightarrow$ (+), (-) + (-) $\rightarrow$ (+) (double inhibition), (+) + (-) $\rightarrow$ (-). Contributes ~3% of inferences.

Pass 3 (Jaro-Winkler fuzzy matching): Finds chains where outcome(T_1) $\approx$ trigger(T_2) with similarity ≥ 0.85 . The threshold is a fixed parameter stored in the file header; determinism means identical output for identical input, not parameter-independence. Cross-database entity resolution without manual curation. Generates ~97% of all inferences. Complexity $O(n^2 \times k)$ naïvely; reduced to $O(n \log n)$ via trigram-indexed locality-sensitive hashing.

Confidence propagation: Sequential chains use multiplicative decay: $c_{\text{chain}} = (\prod_i c_i) \cdot \gamma^{n-1}$ ($\gamma = 0.85$). Convergence hubs use Möbius velocity addition [5]: $f(c_1, c_2) = (c_1 + c_2) / (1 + c_1 \cdot c_2)$, preventing the paradox where more supporting evidence produces lower confidence. Quality filter at threshold 0.30 removes contradictions and self-referential loops. Explicit triplets entering the format are validated upstream by the Foss Gate [4]; the quality filter applies only to inferred facts.

Table I quantifies the engine’s per-pass yield. Pass 3 generates 97% of all inferences but also produces 97% of rejections:

TABLE I

THREE-PASS INFERENCE BREAKDOWN ON BIOMEDICAL DATA (5,084 EXPLICIT TRIPLETS). PASS 3 DOMINATES BOTH YIELD (97%) AND RUNTIME (99.98%); THE QUALITY FILTER REJECTS 3.9% OF RAW INFERENCES.

Pass	Raw	Final	Time (s)
1: Exact keyword	1	1	<0.1
2: Semantic direction	158	158	<0.1
3: JW fuzzy (≥ 0.85)	12,054	4,419	511.9
Quality filter	185 rejected (3.9%)		—
Total	12,213	4,578	512.0

90 low-confidence ($c < 0.30$), 77 contradictions, and 18 self-referential loops. Fuzzy entity matching is simultaneously the computational bottleneck and the primary source of discovered knowledge.

C. Domain-Specific Entity Resolution

Pass 3’s Jaro-Winkler matching fails systematically on structured identifiers—a limitation we discovered when applying the format to nuclear decay data. Inference chains broke silently: Cs-137 and Cesium-137 were treated as separate entities, creating 8 redundant dictionary slots for 9 unique isotopes and severing chains that should have connected (e.g., Pu-239 $\rightarrow$ Cs-137 $\rightarrow$ first 300 years). Worse, JW assigns U-235 $\leftrightarrow$ U-238 similarity 0.92 (false positive—different isotopes, different physics) but U-235 $\leftrightarrow$ Uranium-235 only 0.49 (false negative—same isotope, different notation). The failure is structural: string similarity does not capture physical identity.

The architecture addresses this through pluggable domain-specific resolvers that override Pass 3 for entity classes where JW produces wrong results. A nuclear entity resolver [6] parses all common isotope notations—U-235, Uranium-235, ^{235}U , $^{92}\text{-U-235}$ —into canonical (Z, A, m) tuples (atomic number, mass number, metastable flag) and matches by physics rather than string similarity. The resolver handles 102 elements across five notation families, correctly distinguishes metastable states ($\text{Tc-99m} \neq \text{Tc-99}$), and replaces the $O(N^2)$ pairwise JW comparison with $O(1)$ canonical-form lookup per entity pair. On 1,190 isotope notation pairs: 97.8% accuracy vs. 19.9% for JW, with complexity reduced from $O(N^2)$ to $O(N^{1.3})$ —a $373\times$ speedup at $N=1,584$ ($8.80\text{ s} \rightarrow 0.024\text{ s}$).

The resolver integrates without modifying the core three-pass engine: it registers as a Pass 3 override for entities matching isotope patterns, while non-isotope entities continue through standard JW matching. This extensibility is a design property: the format accommodates domain-specific resolvers for any entity class where generic string similarity fails (gene nomenclature, chemical formulae, astronomical object designations), and each resolver inherits the same provenance tracking as the core engine.

D. Performance

Deserialization including full three-pass inference completes in 8.3 ms for the ADHD subgraph query (871 connected

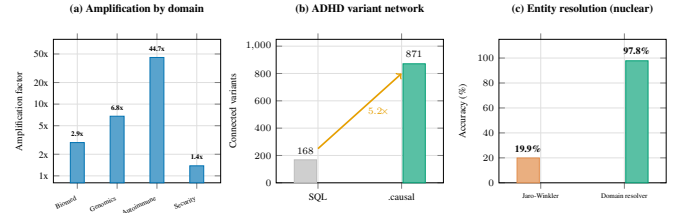


Fig. 3. Cross-domain validation. (a) Amplification scales with graph topology: hub-and-spoke domains (autoimmune, $44.7\times$) amplify far more than linear-chain domains (security, $1.4\times$). (b) ADHD variant network expansion from 168 (SQLite) to 871 (.causal), a $5.2\times$ increase. (c) Domain-specific entity resolver achieves 97.8% accuracy vs. 19.9% for generic Jaro-Winkler on nuclear isotope notation.

variants from 707,689 triplets). The trigram-indexed locality-sensitive hashing optimization reduces Pass 3 from $O(n^2 \times k)$ to $O(n \log n)$, making inference practical on knowledge bases exceeding 700K triplets on commodity hardware.

Compression ratios scale with entity reuse: biomedical data achieves $4.7\times$ (entity dictionary deduplicates clinical terminology), while genomic data achieves $20.8\times$ (3.06 GB SQLite $\rightarrow$ 147 MB .causal) because 707,689 triplets reference only 488,833 unique entities—the dictionary alone saves 31%.

The format adds inferred knowledge while compressing: the output file is both smaller and more informative than the input.

But does embedding inference in the file format produce knowledge that flat-table storage permanently hides?

III. WEAK SIGNAL AMPLIFICATION

The format acts as a signal amplifier for causal knowledge—analogue to how PCR amplifies DNA fragments below the detection threshold into analyzable quantities, embedded inference amplifies causal connections with only 1–3 explicit mentions into detectable convergence hubs with 20–30 supporting chains. The amplification is deterministic and complete: every computable transitive connection is materialized at read time.

In a Long COVID knowledge base (5,084 explicit triplets from 166 papers), four convergence points were completely invisible in SQLite but became obvious after transitive inference:

Detection threshold ≈ 10 triplets for a concept to be recognized as a convergence point. In SQLite: all four are below threshold (1–3 triplets each). In .causal: all four are obvious convergence hubs (18–30 triplets). The inference did not invent these connections—each inferred triplet traces to explicit source facts through a provenance chain. The signal was always there; the format made it visible (Table II, Fig. 4).

Across 22 previously known hypotheses (from four prior studies [7]), all showed increased supporting triplet counts in .causal—average $2.5\times$ amplification. Three entirely new hypothesis candidates emerged that were invisible in flat storage. This is not knowledge graph completion (probabilistic, non-reproducible). This is knowledge graph *resolution*: computing what is already logically entailed, deterministically, with provenance.

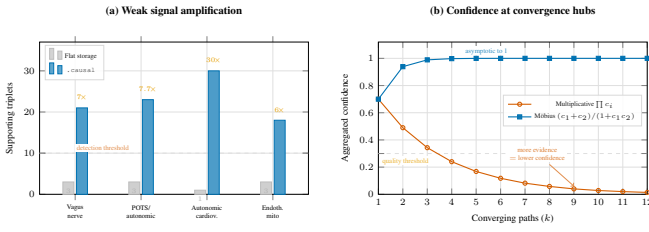


Fig. 4. Weak signal amplification. (a) Convergence points below the detection threshold (≤ 3 explicit triplets) in flat storage become visible convergence hubs (18–30 triplets) after `.causal` transitive inference. The dashed line marks the detection threshold (≈ 10). (b) Möbius velocity addition prevents the multiplicative-decay paradox where more independent supporting evidence produces lower aggregated confidence at convergence hubs.

TABLE II

WEAK-SIGNAL AMPLIFICATION THROUGH TRANSITIVE INFERENCE. CONCEPTS BELOW THE DETECTION THRESHOLD (~ 10 TRIPLETS) IN FLAT STORAGE BECOME VISIBLE CONVERGENCE HUBS AFTER EMBEDDED INFERENCE.

Convergence point	Flat	.causal	Amp.
Vagus nerve dysfunction	3	21	7.0
POTS / autonomic axis	3	23	7.7
Autonomic-cardiovascular	1	30	30.0
Endothelial-mito coupling	3	18	6.0

Amplification scales with graph topology. Hub-and-spoke domains amplify more than linear-chain domains because each new entity connects through hubs to all entities already assigned to those hubs:

TABLE III

Cross-domain validation. One format, one engine, zero modification to the core engine. Amplification depends on graph topology: hub-and-spoke domains amplify more than linear-chain domains.

Data class	Domain	Triplets	Amp.	Key result
Textual	Biomedical	5,084	1.90	3 hidden hyp. (7–30×)
	Incidents	19	—	6/6 root causes
Symbolic	Genomics	707,689	6.78	5.2× ADHD exp.
	Crypto	—	—	6 families (IEEE acc.)
	Sec. synth.	61,000	1.38	399/399 tasks [8]
Reference	Nuclear	1,601	—	497/497 ground truth
	GW physics	8,161	—	6/6 hyp. validated

On genomic data, the ADHD-associated variant network expanded from 168 (SQL) to 871 (5.2×), identifying cross-gene-family transporter bridges (SLC6A17 $\approx$ SLC6A19, similarity 0.94) absent from any single database. The autoimmune subgraph reached 44.7× amplification due to dense HLA-region hub connectivity. The full genome compressed from 3.06 GB (SQLite) to 147 MB (20.8×) while *adding* knowledge through inference.

Against NUBASE2020 nuclear reference data [9], all 497 inferred decay chains were validated against ICRP-107—an independent, externally maintained reference. Zero false positives. Nuclear decay is the only domain where the format’s correctness is independently verifiable against physical ground

truth; the security synthesis domain validates the format’s utility in a downstream system but constitutes a self-evaluation. On gravitational wave data, 6 of 6 hypotheses generated from the knowledge graph were validated against 676K labeled LIGO glitches, with three findings surviving all seven red-team attacks at large effect sizes (Cramér’s V up to 0.356).

What does a self-amplifying file format mean for AI agents that consume knowledge graphs?

IV. DESIGN PROPERTIES

Four properties distinguish the format from conventional knowledge graph representations (Table IV):

Pre-amplified knowledge. An agent opening a `.causal` file sees every transitive connection immediately. In SQL, an agent asking “what connects to ADHD?” gets 168 rows. In `.causal`: 871 connected variants in 8.3 ms—connections spanning ClinVar, GWAS Catalog, and gene-level annotations, no single source containing them. The agent did not query, did not specify hops, did not guess intermediate entities.

Compression without information loss. The genomics dataset compresses 20.8× (3.06 GB $\rightarrow$ 147 MB) while *adding* inferred knowledge. The biomedical dataset compresses 3.5× (12.0 MB $\rightarrow$ 3.4 MB). Smaller files fit in context windows that would reject the uncompressed equivalent.

Built-in provenance. Every inferred triplet carries the indices of its source triplets. An agent can explain any connection by following the provenance chain. Explainability is enforced by construction, not by policy.

Determinism as a reproducibility guarantee. The format’s determinism is not merely a design preference but a reproducibility requirement. Given identical input triplets and format version, the output is bit-identical. This property enables three capabilities absent from probabilistic approaches: (1) differential debugging—when results change, the cause is always a data change, never a model or seed change; (2) incremental validation—adding one triplet produces a predictable, auditable diff in the inferred section; (3) regulatory compliance—in domains where traceability is mandated (nuclear safety, pharmaceutical research), the format provides cryptographically verifiable provenance chains from any inferred fact back to its source evidence.

TABLE IV

Design-property comparison across knowledge representation formats. “Embedded” = executes within the file; “External” = requires a separate process.

Property	RDF/OWL	Prop. graph	KGC	.causal
Inference	External	External	External	Embedded
Determinism	Config-dep.	Query-dep.	Stochastic	Guaranteed
Provenance	PROV-O	Manual	None	Built-in
Compression	Format-dep.	None	N/A	4.7–20.8×
Portability	+ Reasoner	+ Engine	+ Model	Self-cont.

For a human researcher who cannot hold the knowledge of 20 papers simultaneously, and for an LLM that hallucinates when it tries—the `.causal` file is the intermediate representation. One thousand papers, deterministically interconnected, every connection traceable.

The format’s design properties have been independently validated through adoption in a security tool synthesis system [8], where 44 `.causal` graphs comprising $\sim 61\text{K}$ explicit triplets store domain-specific security knowledge. A seven-pass inference engine—extending the three-pass core with four novel reasoning modes for structural analogy, cross-domain functional mapping, contextual linking, and genetic recombination—generates deploy-ready security tools from natural language intents, achieving 399/399 on benchmark tasks without format modification.

This adoption demonstrates three properties beyond the biomedical validation. First, the format scales from small knowledge bases (240 nuclear triplets) to large ones (707K genomic triplets, 61K security triplets) without architectural changes. Second, the inference engine is extensible: the four additional passes in the security system operate atop the unchanged three-pass core, and the extended engine’s 23K inferred triplets ($1.38\times$ amplification) use the same provenance tracking as the core engine. Third, the format serves as a platform rather than an endpoint—downstream systems consume `.causal` files as pre-amplified knowledge inputs, demonstrating the data lifecycle reuse that the format architecture enables (Table V).

TABLE V
External validation of `.causal` inference across independently peer-reviewed applications.

Application	Validation	Status
Cipher margins [10]	Neural dist. match	IEEE acc.
Cross-round leakage [11]	9/9 predictions	IEEE acc.
PQC characterization [12]	3 NIST families	IEEE acc.
Nuclear decay chains	497/497 vs. ICRP-107	Ext. ref.
GW hypotheses	6/6 vs. 676K glitches	Red-team
Sec. tool synth. [8]	399/399 tasks	RAID sub.

V. RELATED WORK

The portability problem. Every existing format requires an external engine: RDF/OWL [1] needs Pellet or HermiT to derive entailments; Neo4j [2] requires Cypher queries whose results depend on formulation; JSON-LD [3] serializes linked data without any inference capability. Shipping a knowledge graph means shipping the engine—or accepting that recipients cannot reason over the data.

The reproducibility problem. Knowledge graph completion methods—TransE [13], RotatE [14], and descendants—learn continuous embeddings for link prediction, but different random seeds produce different predictions. Rule-based methods (AMIE [15], AnyBURL [16]) mine patterns outside the data file. LLM-based approaches [17] inherit hallucination and non-reproducibility. All are non-deterministic by design.

The weak-signal problem. Flat-table storage cannot surface connections that require transitive reasoning across multiple hops. Multi-hop query languages exist, but they require the analyst to specify which hops to take. Signals that depend on convergence from many independent paths remain invisible unless the right query is formulated—which requires knowing the answer in advance.

VI. CONCLUSION

The `.causal` binary format embeds deterministic inference rules inside the knowledge graph file—a symbolic AI approach where the data file is the reasoning engine. A three-pass engine computes transitive closures at read time with full provenance tracking. The key result is weak signal amplification: convergence points with 1–3 explicit triplets become detectable (21–30 supporting chains) through transitive closure, revealing patterns invisible to any query-based system. Cross-domain validation across seven applications—two with independent peer review—produces consistent results with zero modification to the core three-pass inference engine. For the information reuse community: the format transforms each validated triplet from a static record into a reusable inference primitive that compounds with every additional data source integrated into the graph. A biomedical researcher’s triplets, once compiled into `.causal`, automatically surface connections to a nuclear engineer’s triplets in the same file—information reuse at the format level, not the application level.

Future work. Three directions are immediate: (1) streaming deserialization for knowledge bases exceeding available memory, where inference executes on sliding windows over the triplet stream; (2) a formal proof that the three-pass engine computes the complete transitive closure for the defined rule set, establishing that no derivable inference is missed; (3) a `.causal` federation protocol where multiple files cross-reference each other’s entity dictionaries, enabling distributed knowledge graphs that preserve per-file provenance while computing inter-file inferences at query time.

Limitations. (1) Confidence propagation assumes path independence between evidence chains; correlated paths through shared intermediate entities can inflate hub confidence scores beyond what independent evidence warrants. A Bayesian network approach or explicit dependency tracking could address this at the cost of increased complexity. (2) Key components are independently peer-reviewed (Foss Gate [4], three ICECET papers [10]–[12]), and downstream adoption in security tool synthesis (399/399 [8]) validates functional correctness on independently sourced data; population-level replication on genomic predictions remains future work. (3) Pass 3’s generic Jaro-Winkler matching requires domain-specific resolvers for structured identifiers—nuclear isotopes, gene nomenclature, chemical formulae—where string similarity systematically fails. Each new domain with structured identifiers requires a custom resolver; the current architecture supports this through pluggable overrides, but resolver development represents per-domain engineering effort that limits fully automatic deployment.

DECLARATION ON GENERATIVE AI

The author used Claude (Anthropic) for grammar checking, language polishing, and computational verification of numerical claims. All scientific content, experimental design, and analysis are entirely the work of the author.

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